How to run the model

DVR and D-STATCOM with hybrid PV–wind generation on the IEEE 13-bus feeder

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The only file needed: `Simulink_model/IEEE13_DVR_DSTATCOM_PV_Wind.slx`

Every step of the study (base case, D-STATCOM, DVR, PV and wind, all together) runs from this one model with the Simulink **Run** button. There is no script to run and nothing to add to the MATLAB path: the feeder data and the four controllers are stored inside the model. Every value you may want to change is a block that you double-click.

1. What you need

- MATLAB **R2024b** with **Simulink** and **Simscape Electrical** (Specialized Power Systems).
- For MATLAB R2022a or R2022b, open the copy in `MATLAB_R2022/Simulink_model/` instead. It is the same model saved in the older format.

2. Run it

1. Open `IEEE13_DVR_DSTATCOM_PV_Wind.slx` (double-click it, or open it from MATLAB).
2. Press **Run** on the Simulation tab. The stop time is 1 s and a run takes about a minute; the first run takes longer while Simulink compiles the model.
3. Five scope windows open and fill in as the model runs.
4. When the run ends, read the displays to the right of the **Measurements** block, below the feeder.

MATLAB may print *"Matrix is close to singular or badly scaled"* when the run starts. The original IEEE 13-bus benchmark model prints the same warning; it does not stop the run.

3. Saved scenario

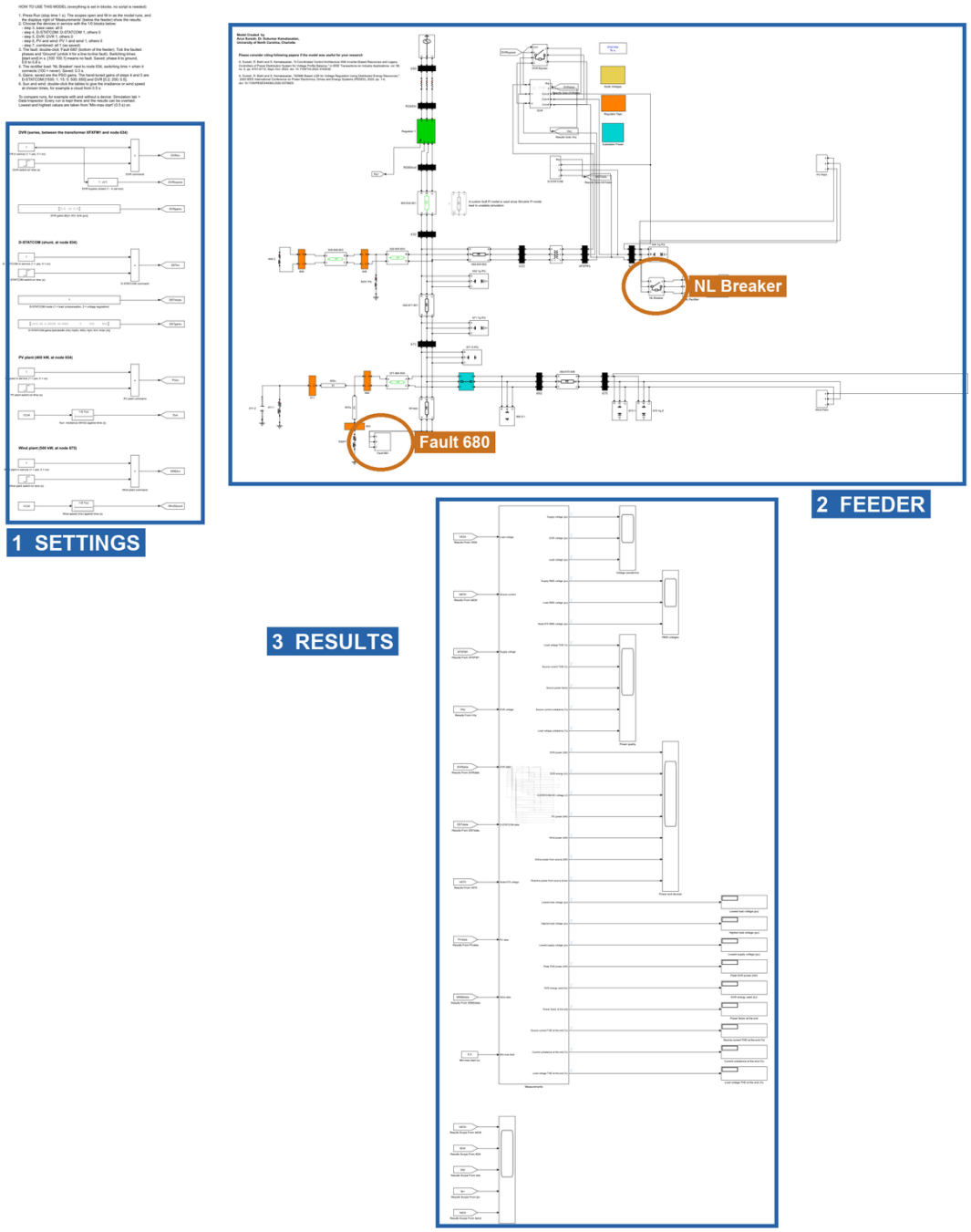
As saved, the model runs the combined case of step 7 with this scenario:

Time	What happens
0.2 s	PV and wind plants connect
0.3 s	100 kW rectifier (non-linear) load connects at node 634
0.4 s	DVR and D-STATCOM start
0.6–0.8 s	Single line-to-ground fault on phase A at node 680
1.0 s	End of the run

4. Where things are in the model

The model has three areas. Only area 1 and the two blocks circled in area 2 are meant to be changed.

Area	What is there
1 Settings (top left)	A note “HOW TO USE THIS MODEL” and, below it, the settings of the DVR, D-STATCOM, PV plant and wind plant.
2 Feeder	The IEEE 13-bus feeder with the four devices. The fault block **Fault 680** and the rectifier switch **NL Breaker** are here.
3 Results (below the feeder)	The **Measurements** subsystem, five scopes and the displays.



The whole model. Blue boxes: the three areas. Orange: the two blocks in the feeder that you may change.

5. What you can change

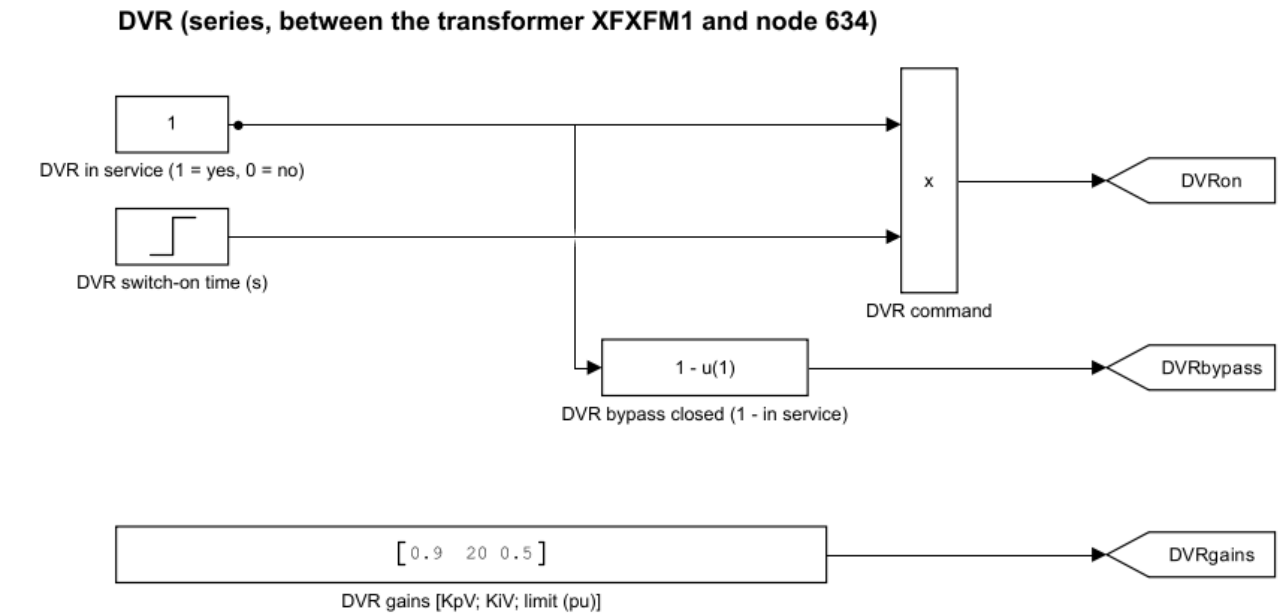
To change a value: **double-click the block**, edit the field named below, click **OK**, and press **Run** again. Block names are written in the model exactly as in the first column.

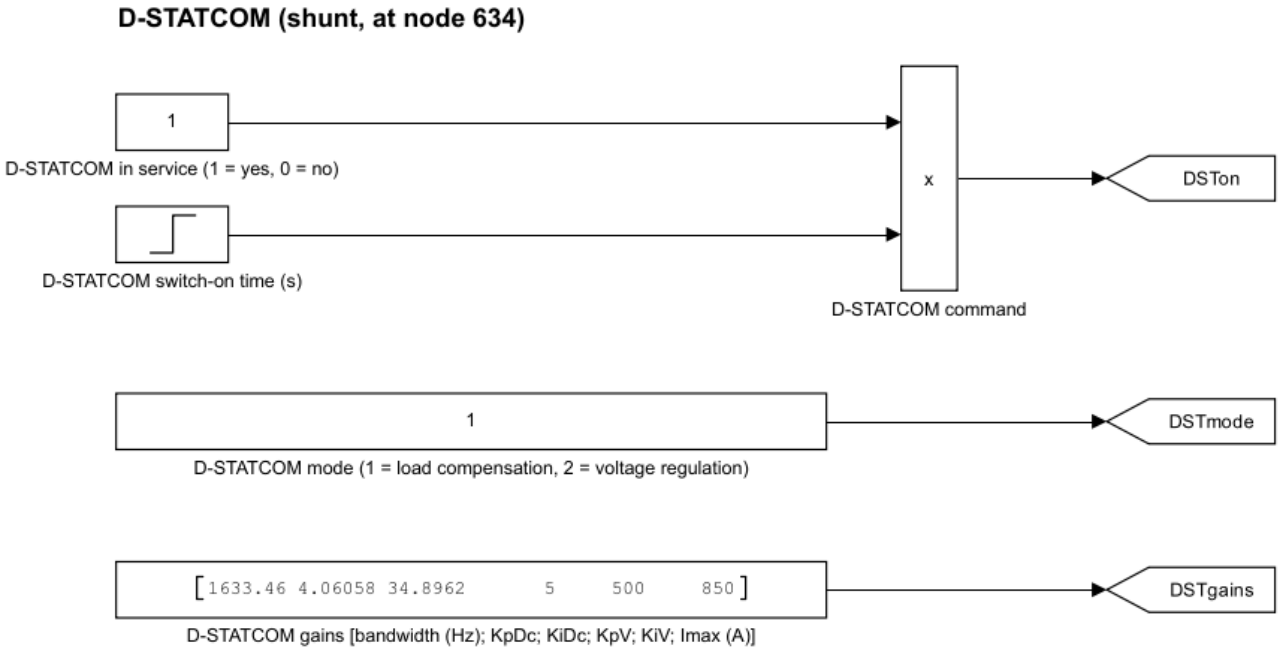
Block	Field to edit	Saved value	What it means
Settings area			
DVR in service (1 = yes, 0 = no)	Constant value	1	0 takes the DVR out: a bypass breaker closes around it
DVR switch-on time (s)	Step time	0.4	When the DVR starts injecting
DVR gains [KpV; KiV; limit (pu)]	Constant value	[0.9;20;0.5]	PSO gains. Hand-tuned: [0.2;200;0.5]. Limit = largest injection per phase
D-STATCOM in service (1 = yes, 0 = no)	Constant value	1	0 keeps its breaker open
D-STATCOM switch-on time (s)	Step time	0.4	When its breaker closes and control starts
D-STATCOM mode (1 = load compensation, 2 = voltage regulation)	Constant value	1	1: corrects power factor, unbalance and harmonics of the load current. 2: supports the voltage
D-STATCOM gains [bandwidth (Hz); KpDc; KiDc; KpV; KiV; Imax (A)]	Constant value	[1633.46; 4.06058; 34.8962; 5; 500; 850]	PSO gains. Hand-tuned: [1500;1;15;5;500;850]
PV plant in service (1 = yes, 0 = no)	Constant value	1	0 keeps its breaker open
PV plant switch-on time (s)	Step time	0.2	When the PV plant connects
Sun: irradiance (W/m2) against time (s)	Breakpoints (times) and Table data (values)	[0 100] and [1000 1000]	Irradiance at each time; saved: 1000 W/m² all the time
Wind plant in service (1 = yes, 0 = no)	Constant value	1	0 keeps its breaker open
Wind plant switch-on time (s)	Step time	0.2	When the wind plant connects
Wind speed (m/s) against time (s)	Breakpoints (times) and Table data (values)	[0 100] and [12 12]	Wind speed at each time; saved: 12 m/s (rated) all the time
Feeder			
Fault 680	Phase A, Phase B, Phase C, Ground (tick boxes)	A and Ground	Fault type: see section 7
Fault 680	Switching times (s)	[0.6 0.8]	[start end] of the fault
Fault 680	Fault resistance Ron (Ohm)	0.01	Larger value: milder fault
NL Breaker	Switching times (s)	[0.3]	When the rectifier load connects; 100 = never
Results and toolbar			
Min-max start (s)	Constant value	0.5	Lowest and highest values are taken from this time on
Stop time (Simulation tab)		1.0	Length of the run in seconds

A time later than the stop time, such as 100, means “never” within the run.

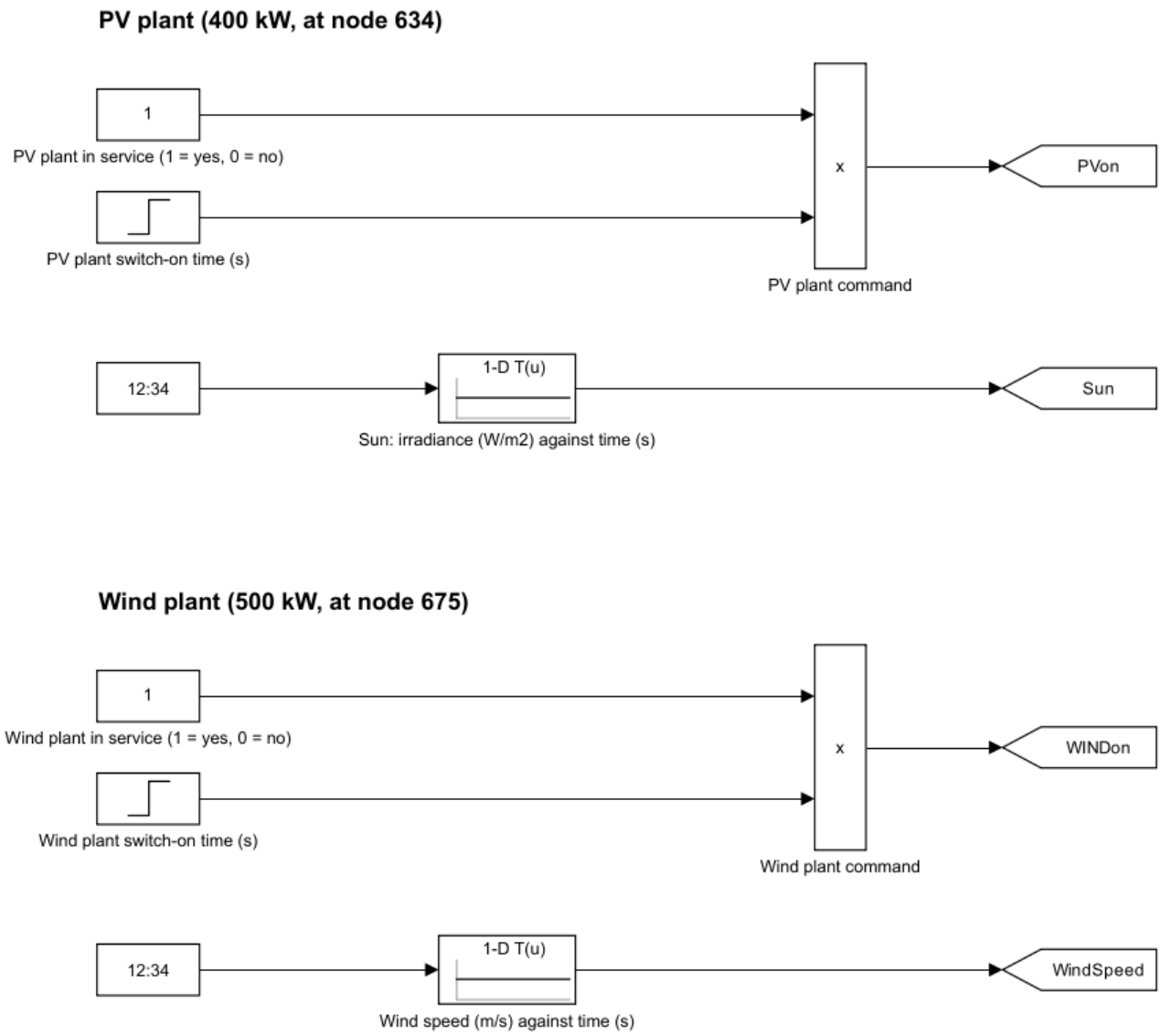
The settings area

Each device has a 1/0 block, a switch-on time and, for the DVR and D-STATCOM, its gains. The blocks on the right (DVRon, Sun and so on) pass the values to the devices; they are not to be changed.





Settings of the DVR and the D-STATCOM.



Settings of the PV and wind plants.

6. Running each step of the study

Each step is the same model with some devices out of service. Set the 1/0 blocks as below and keep the saved fault. Steps 4 and 5 were done with the hand-tuned gains: type them into the gain blocks to get the numbers in the last column.

Step	Set to 1 (others 0)	Gains	What the results show
3 Base case	none	–	Load voltage sags to 0.61 pu on phase A and swells to 1.11 pu on B and C; power factor 0.86; current unbalance 7.1 %
4 D-STATCOM	D-STATCOM	hand-tuned	Power factor 1.00; current unbalance 0.7 %; current THD 2.4–2.8 %
5 DVR	DVR	hand-tuned	Supply sags to 0.59 pu while the load stays at 1.00 pu; peak DVR power 121 kW
6 PV and wind	PV plant, wind plant	–	PV about 400 kW, wind about 500 kW; node 675 falls to 0.21 pu during the fault
7 Combined	all four (as saved)	PSO (as saved)	Load within 1 % (0.996–1.01 pu); power factor 1.00; peak DVR power 47 kW

7. Choosing the fault

Double-click **Fault 680**. Tick the faulted phases and *Ground*, and give *Switching times (s)* as [start end]. Keep the start after the devices switch on (0.4 s) and the end before the stop time.

Fault	Phase A	Phase B	Phase C	Ground
Single line-to-ground, phase A (saved)	ticked			ticked
Double line-to-ground, phases B–C		ticked	ticked	ticked
Line-to-line, phases B–C		ticked	ticked	
Three-phase-to-ground	ticked	ticked	ticked	ticked
No fault	Switching times (s) = [100 100.1]			

8. Examples of changes

To try	Change	Where to look
A cloud over the PV plant	**Sun** table: Breakpoints [0 0.5 0.55 100], Table data [1000 1000 300 300] (1000 to 300 W/m² between 0.5 and 0.55 s)	Power and devices scope: PV power
Less wind	**Wind speed** table: Breakpoints [0 0.5 0.6 100], Table data [12 12 9 9]. The turbine slows gradually; a longer stop time (2 s) shows more of it	Power and devices scope: wind power
The DVR starts late	**DVR switch-on time (s)**: 0.7	RMS voltages scope: the load sags at 0.6 s until the DVR starts
Without the rectifier load	**NL Breaker**, Switching times (s): 100	Power quality scope: THD
Only the rectifier, no fault	**Fault 680**, Switching times (s): [100 100.1]	Power quality scope
A milder fault	**Fault 680**, Fault resistance Ron (Ohm): 1	RMS voltages scope

In the Sun and Wind speed tables, the breakpoints (times) must increase, and the two rows must have the same number of values.

9. Reading the results

Scopes (open when the run starts)

- **Voltage waveforms**: supply, DVR and load voltage (pu)
- **RMS voltages**: supply, load and node 675 (pu)
- **Power quality**: voltage and current THD, power factor, current and voltage unbalance
- **Power and devices**: DVR power and energy, D-STATCOM DC voltage, PV and wind power, power from the source
- **Currents**: source, load, D-STATCOM, PV and wind currents (A)

Displays (values at the end of the run)

- Lowest and highest load voltage, lowest supply voltage (pu)
- Peak DVR power (kW) and DVR energy used (kJ)
- Power factor, source current THD, current unbalance and load voltage THD at the end

Comparing runs

Every run is kept in the **Data Inspector** (Simulation tab › Data Inspector). Each run is listed separately; tick the same signal, for example *Load RMS voltage (pu)*, in two runs to draw them on one plot. A useful comparison: run once with all four devices at 0 and once with all at 1, and overlay the load RMS voltage.

10. Good to know

- **To go back to the saved settings**, close the model without saving. To keep a scenario, save a copy with Save As under another name.
- Change values only in the settings area, in **Fault 680** and in **NL Breaker**. Leave the device subsystems (DVR, D-STATCOM, PV Plant, Wind Plant) and the Measurements block as they are.
- Keep the switch-on times before the fault, and **Min-max start (s)** after the switch-on times, so that start-up transients do not count as the lowest or highest voltage.
- Simulink makes a folder `slprj` and a file ending in `.slxc` next to the model when it runs. They are temporary and can be deleted.
- If the model does not open in an older MATLAB, use the R2022 copy. If Simulink reports missing blocks, Simscape Electrical is not installed.
- The step-by-step models and the scripts in the project folder are only needed for the optimisation and machine-learning work. The gains found by the optimisation (PSO) are already saved in this model.